

STEP-5B exhaustiveness decision support: the T-031 operator-decision dossier

TECT verification-first repository · claim B5-BEYOND-LAYER-BOUND

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1. Purpose and scope

T-031 is the remaining DECISION layer of STEP-5B: the budget COMPARISON is machine-closed and the analytic inputs are published; what remains are operator rulings on class definitions and gate/tier consequences. This note is the dossier for those rulings. **Scope:** enumeration + evidence state + options. **Not claimed:** no new mathematics; nothing is enacted here.

2. What is already closed (no decision needed)

(i) Budget comparison: $K(n_{\text{pack}}) \leq 113$ (rectangle bound) vs measured $K = 107$, against $K_{\text{budget}} = 5972/927/221$ – margins $\times 55.6/ \times 8.8/ \times 2.1$ ($\times 2.6$ with the J-refined endpoint); official threshold $n_{\text{adm}} \lesssim 1.59 \times 10^5$ vs $n_{\text{pack}} \approx 44\text{--}49$ ($\times 3200$). (ii) The rectangle prefactor: PUBLISHED (STEP-5B-Rectangle-T6-260612). (iii) Lattice-class additive energy: PUBLISHED, T7 CONFIRMED (DR2-Lattice-T7-NTstandard-260612; Lemma NT pinned in-bundle, 2,644,976-sum exhaustive applied check). (iv) The tadpole channel: gates R-U6-1/R-U6-2 DISCHARGED 2026-06-12 – the third-cumulant threat is SUNSET-ONLY, and the endpoint joint is certified $\times 1.040\text{--}\times 1.082 > 1$ on $\mathcal{W}_{\text{SC}}$ (SC-SCOPE-T5-260612). (v) RES-4 layer ratio $\rho(I) \geq 2.58$ (monotone, 201 nodes).

3. Decision item D1: H-ADM-COH discharge for the lattice class

Question. Is the admissibility hypothesis H-ADM-COH (coherence-derived competitor class: registered shells/shell-unions, $\theta_{\min}$ separation, $n \leq n_{\text{pack}}$) DISCHARGED – i.e. no longer a hypothesis but a derived constraint – for the lattice competitor class?

Evidence state (per GATES.md R-028 + the 2026-06-12 closures). R-027 (weighted Lemma A, T7) + R-026 (lattice $T' \ll_{\varepsilon} R^{\varepsilon}$, now T7-NTstandard with the NT pin in-bundle) give the weighted additive-energy bound with $K \sim R^{\varepsilon}$ for the unrestricted LATTICE class WITHOUT the separation input; the finite margin $K_{\text{adm}} = 1 + T'(Q) \leq 8 + 4\sqrt{14}\sqrt{n}$ is machine-verified at $\times 10.6\text{--}\times 15.8$ and growing. What R-028 left as the discharge residuals: (a) the subpolynomial- K sufficiency over the admissible R -range – NOW PINNED (the exhaustive 2.6M-sum check bounds every applied instance; enumerated competitors close exactly at $K_{\text{floor}} \leq 12$); (b) the modeling residual (carrier-richness $\chi(P) \lesssim T'$ link) – unchanged, an operator judgment by design.

Options.

- D1-A. DISCHARGE for the lattice class (H-ADM-COH becomes a derived constraint there; the amended-class qualifier in B5's scope narrows to the non-lattice remainder). Ledger effect: B5 scope string updated; B1 unaffected (rests on H-LAYER alone); DR2-SHARE handling moves to D2.
- D1-B. HOLD (status quo): H-ADM-COH stays a named class definition. No ledger change; T-031 stays open on this item.

4. Decision item D2: the DR2-SHARE gate

Question. GATES.md carries DR2-SHARE as OPEN “pending operator discharge” (the unrestricted/arbitrary- Q additive-energy share). With the synthesis-layer finding that arbitrary- Q DR-2 is NOT load-bearing for the C_{full} theorem (Lemma 2 caps $T' \leq 10$ in-class) and the lattice class now T7-confirmed, should DR2-SHARE be (A) re-scoped to the explicitly non-load-bearing frontier track T-030, or (B) kept OPEN as-is?

Note. Option A is bookkeeping (the mathematical content of T-030 is unchanged); it prevents the OPEN gate from being misread as a hole in the published theorem.

5. Decision item D3: the B5 claim tier

Question. B5 is T5 PINNED-CLOSURE. The T5-DOSSIER’s original “below T6” rationale cited: (i) T4-grade lemmas in the chain; (ii) the endpoint at $\times 2.1$ (now hardened $\times 2.6$); (iii) SC-SCOPE marginal (now LIFTED@THIN-CERTIFIED with the certified $\times 1.040$ joint and the tadpole struck). Post-2026-06-12, the surviving sub-T6 elements are: the endpoint thinness (structural, sunset-capped); the H-ADM-COH class pin (D1); and the estimate-grade third-cumulant inflation basis noted in GATES.

Options.

- D3-A. Re-tier B5 to T6 PROVED CONDITIONAL on the explicit hypothesis set {H-ADM-COH (if D1-B) or non-lattice remainder (if D1-A); endpoint thinness acceptance}. Requires: a fresh devil’s-advocate pass on the claim card + the atomic flip set.
- D3-B. Stay T5 PINNED-CLOSURE (status quo) until the sunset endpoint is hardened beyond thin (e.g. a proved $K_{\text{floor}} \leq 0.52 T'$ floor or a tighter sunset accounting). No ledger change.

The synthesis layer’s anti-tier-shopping rule is unaffected by either choice (units keep their own tiers).

6. What NO decision here changes

The published main theorem (T7-SCOPE $_{C_{\text{full}}}$ given A1, window $\mathcal{R}$) and the B1/B2 canonical states are carried by the head package and are INDEPENDENT of D1–D3; the five published bundles and the synthesis layer are frozen artifacts. D1–D3 affect only B5’s own scope string, the GATES.md rows, and the claim-tier bookkeeping.

7. Devil’s-advocate / self-adversarial review (CLAUDE.md 6.3)

- (α) “*A decision dossier can frame options to favor promotion.*” ADDRESSED: every option includes the status-quo branch with “no ledger change”; the D3 surviving sub-T6 elements are enumerated from the original dossier’s own rationale, not re-argued; nothing is enacted.
- (β) “*The D1 residual (b) ($\chi(P) \lesssim T'$) might be load-bearing, not a judgment.*” VALID with mitigation: per the B5 record (R-026/R-027 verdicts) this link was explicitly classified an OPERATOR-DECISION; the dossier preserves that classification and does not smuggle it into the evidence column.
- (γ) “*Quoting the hardened endpoint ($\times 2.6$) without its conditions.*” DISMISSED: the J-refinement conditions (conservative criterion band) are in the published STEP-5B bundle’s expected outputs; the conservative end $\times 2.1$ is quoted alongside throughout.

Quantitative sanity check (mandatory). All numbers quoted here are published-bundle values (margins $\times 55.6 / \times 8.8 / \times 2.1$ – 2.6 ; $K \leq 113$ vs 107; $\times 10.6$ – $\times 15.8$ R-028 margins; joint $\times 1.040$ – $\times 1.082$; $\rho \geq 2.58$; threshold 1.59×10^5 vs $n_{\text{pack}} \approx 49$); no new computation enters a decision input.

8. Result footer

Result ID: B5-BEYOND-LAYER-BOUND / STEP-5B
exhaustiveness decision support (T-031)
Precise statement: DECISION DOSSIER: three operator items --
D1 H-ADM-COH lattice-class discharge
(evidence: R-026 T7-NTstandard + R-027 +
R-028 margins x10.6-15.8 + 2.6M exhaustive
check; residual (b) $\chi(P) < \sim T'$ stays a
judgment); D2 DR2-SHARE re-scope to the
non-load-bearing frontier (bookkeeping);
D3 B5 tier (T6-conditional re-tier vs stay
T5). Nothing enacted.
Scope: decision layer only; no new math, no flips.
Evidence grade: DOSSIER over published bundles + GATES.md
records.
Reproduction command: none new (cites published bundle manifests).
Falsifier: any quoted number diverging from its bundle
source.
Tier before / after: no change (B5 stays T5 pending D3).
No-overclaim statement: options are presented with status-quo
branches; the operator-decision
classifications of the B5 record are
preserved unchanged.
Next required action: operator verdicts on D1 / D2 / D3 (each a
single ruling); then enact the chosen
options as an atomic flip set.